

Rithish S

Software Engineering Intern | Frontend Architecture (React/Next.js), Interactive WebGL/Canvas & GenAI Systems

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EDUCATION

Bannari Amman Institute of Technology
Bachelor of Engineering in Computer Science and Engineering
CGPA: **8.28/10**

Sathyamangalam, Tamil Nadu
Aug. 2024 – May 2028

TECHNICAL SKILLS

Frontend & UI: React, Next.js, TypeScript, JavaScript (ES6+), HTML5 Canvas, WebGL, Tailwind CSS, Redux Toolkit
Languages & Core CS: Python, C++, Java, SQL, Object-Oriented Design (OOP), Data Structures & Algorithms
Backend & APIs: Node.js, Fastify, Express.js, REST APIs, Microservices, Redis Caching, k6 Performance Testing
AI & Cloud Systems: Generative AI, RAG Pipelines, Vector Search (Pinecone), AWS (EC2, S3), Docker, Linux (POSIX), Git
Certifications: AWS Cloud Practitioner Essentials, Oracle Cloud Foundations Associate, Cisco Linux & OS

EXPERIENCE

Web Developer <i>Students Special Group</i> - Engineered modular full-stack web applications using React, Node.js, and MySQL, architecting reusable UI component hierarchies and responsive state management across institutional platforms. - Designed and integrated high-throughput REST APIs and normalized relational schemas for a Faculty Dashboard, ensuring sub-second response times and transactional consistency. - Mentored 50+ students in modern JavaScript (ES6+), React component design patterns, Git workflows, and front-end performance best practices.	Feb. 2025 – Dec. 2025 <i>Student Organization</i>
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FEATURED PROJECTS

Plantera | Interactive AI & WebGL Spatial Platform | *React, Mapbox GL / WebGL, Python, Node.js, Tailwind CSS* [GitHub](#)

- Architected an interactive geospatial monitoring application using **React and Mapbox GL**, utilizing **hardware-accelerated WebGL vector rendering** and **HTML5 Canvas** overlays for real-time visualization.
- Built modular, reusable UI components and custom interactive controls, achieving fluid 60fps pan/zoom interactions and responsive client-side state synchronization across diverse viewports.
- Integrated backend machine learning prediction pipelines with REST APIs; awarded **1st place among 120+ teams** at the IEEE DevSpark Hackathon.

PatentIQ | AI Prior-Art & Novelty Engine | *Node.js, Fastify, Python, PostgreSQL, Pinecone, Ollama, Prisma* [GitHub](#)

- Engineered an AI-native novelty search engine using Pinecone vector similarity search and local LLM embeddings (Qwen2.5) for automated claim-overlap analysis and technical risk scoring.
- Architected a high-throughput Fastify backend utilizing Clean Architecture, dependency injection, and Prisma/PostgreSQL, optimizing API response latencies for complex analytical queries.
- Implemented JWT authentication, Zod schema validation, and secure transactional pipelines handling multi-step analysis workflows.

OPEN SOURCE

ESLint | *JavaScript, AST, Static Analysis, IEEE 754, CI/CD* [PR #21218](#)

- Contributed a merged fix to ESLint's **no-loss-of-precision** rule, distinguishing genuine zero literals from non-zero decimals that underflow under IEEE 754 binary64 floating-point specifications.
- Authored a comprehensive regression test suite covering underflow, subnormal values, signed zero, and non-decimal literals; passed **43 automated CI checks** prior to production merge.

ACHIEVEMENTS

- 1200+ DSA Problems Solved** across LeetCode, CodeChef, and GeeksforGeeks, spanning dynamic programming, graphs, trees, and algorithmic complexity.
- IEEE DevSpark Hackathon Winner** – Secured **1st place among 120+ teams** in Full Stack Development by building Plantera, an AI-powered interactive environmental monitoring platform.
- BIT Hackathon Winner** – Secured **1st place among 200+ teams** by architecting a full-stack Leave Management System under hackathon time constraints.
- Official Cloud Certifications** – Certified in **AWS Cloud Practitioner Essentials** and **Oracle Cloud Infrastructure Foundations Associate** (2025).